SMART RESTAURANT MANAGEMENT SYSTEM

Database Schema

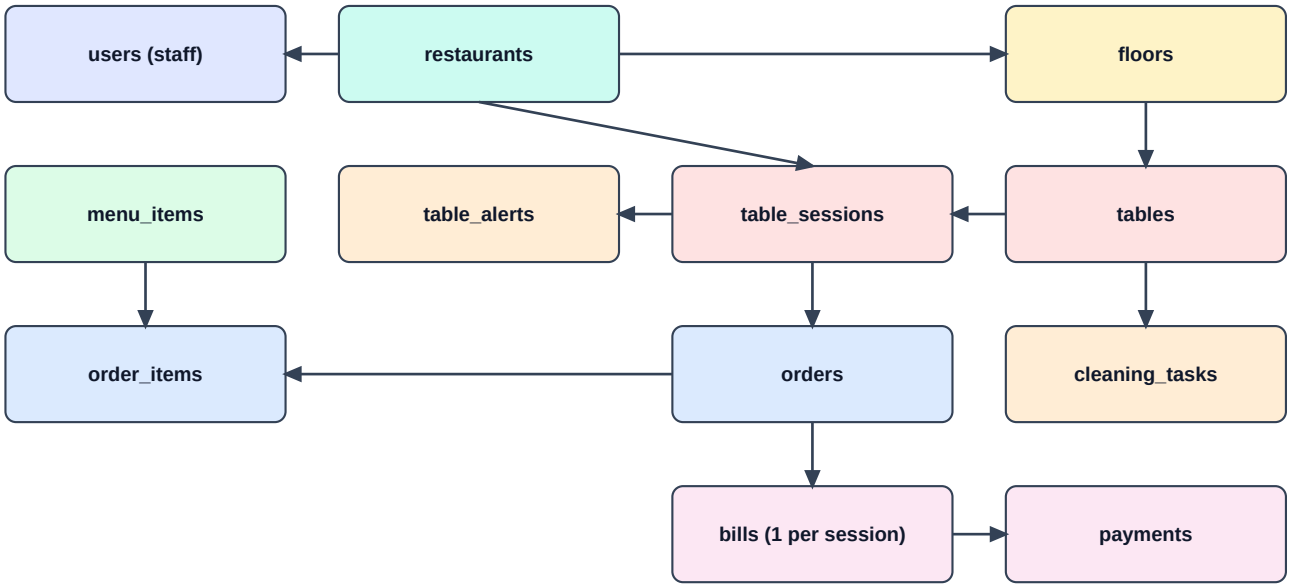
PostgreSQL tables, relations, enums and indexes

DOCUMENT ID	SRMS-04
VERSION	1.0
DATE	06 November 2026
STATUS	Draft for review
PHASE	Phase 1: event-driven core | Phase 2: AI camera detection
PURPOSE	Complete data model for Phase 1 and the Phase 2 AI camera extension.

1. Design Principles

- **Database:** PostgreSQL 15+. Primary keys are UUIDs; timestamps are `timestamptz` in UTC.
- **Multi-tenant ready:** every business table carries `restaurant_id` so multiple outlets can share one database.
- **Snapshots:** order items store name and price at order time so later menu edits never change history.
- **Money:** `numeric(12,2)`, never floating point. Currency is INR by default.
- **Soft delete:** menu and staff records use `is_active` or `deleted_at`; financial records are never deleted.
- **Audit:** status changes and sensitive actions are written to log tables.
- **Single source of truth:** table status lives in `tables.status` and is changed only by the state engine, with every change logged.

2. Entity Relationship Summary



Reading the diagram: a restaurant has floors and staff; a floor has tables; a table has many sessions over time; each session has orders (with items from the menu), one bill, and the bill has payments. Alerts and cleaning tasks also hang off the table and session.

3. Enumerations

Enum	Values
user_role	OWNER, MANAGER, WAITER, KITCHEN, CLEANER
table_status	AVAILABLE, RESERVED, OCCUPIED, PAYMENT_PENDING, NEEDS_CLEANING, CLEANING_IN_PROGRESS, OUT_OF_SERVICE
session_status	ACTIVE, CLOSED, EXPIRED
order_status	PLACED, CONFIRMED, IN_PROGRESS, READY, SERVED, CANCELLED
item_status	PENDING, ACCEPTED, PREPARING, READY, SERVED, CANCELLED
bill_status	OPEN, PAID, PARTIALLY_PAID, VOID
payment_method	UPI, CARD, WALLET, CASH
payment_status	CREATED, PENDING, SUCCESS, FAILED, REFUNDED, PARTIALLY_REFUNDED

Enum	Values
alert_type	CALL_WAITER, REQUEST_BILL, ORDER_DELAYED, PAYMENT_FAILED, LONG_WAIT, UNATTENDED, LONG_STAY, HELP, AI_SUGGESTION
alert_status	OPEN, ACKNOWLEDGED, RESOLVED
task_status	PENDING, IN_PROGRESS, DONE, CANCELLED
actor_type	CUSTOMER, STAFF, SYSTEM, AI

4. Table Definitions

4.1 Organisation and Staff

restaurants

One row per outlet.

Column	Type	Constraints / Notes
id	uuid PK	default gen_random_uuid()
name	varchar(150)	not null
slug	varchar(80)	unique; used in URLs
address, phone	text, varchar(20)	
gstin	varchar(15)	nullable
currency	char(3)	default INR
timezone	varchar(50)	default Asia/Kolkata
settings	jsonb	taxes, service charge, thresholds, session timeouts
created_at	timestamptz	default now()

users

Staff accounts (guests are not users).

Column	Type	Constraints / Notes
id	uuid PK	
restaurant_id	uuid FK	references restaurants(id)
name	varchar(120)	not null
email	varchar(160)	unique per restaurant; nullable for PIN-only staff
phone	varchar(20)	
password_hash	text	argon2/bcrypt; nullable
pin_hash	text	hashed staff PIN
role	user_role	not null
is_active	boolean	default true
last_login_at	timestamptz	
created_at	timestamptz	

staff_shifts

Who is on duty and which tables they cover.

Column	Type	Constraints / Notes
id	uuid PK	
user_id	uuid FK	references users(id)
floor_id	uuid FK	nullable
assigned_table_ids	uuid[]	tables this waiter covers
starts_at, ends_at	timestamptz	

4.2 Floor and Tables

floors

Zones or floors (Indoor, Terrace, Family section).

Column	Type	Constraints / Notes
id	uuid PK	
restaurant_id	uuid FK	
name	varchar(80)	
sort_order	int	default 0

tables

Physical tables and their live status.

Column	Type	Constraints / Notes
id	uuid PK	
restaurant_id	uuid FK	
floor_id	uuid FK	
label	varchar(20)	e.g. T04; unique per restaurant
capacity	smallint	seats
shape	varchar(12)	round, square, rect
pos_x, pos_y, width, height	numeric	floor-map layout
status	table_status	default AVAILABLE; changed only by state engine
status_changed_at	timestamptz	drives timers
current_session_id	uuid FK null	active session if any
qr_token	varchar(64)	unique, random, rotatable
assigned_waiter_id	uuid FK null	default waiter
is_active	boolean	default true

4.3 Sessions

table_sessions

One occupancy period of a table.

Column	Type	Constraints / Notes
id	uuid PK	
restaurant_id	uuid FK	
table_id	uuid FK	indexed
status	session_status	default ACTIVE
opened_by	actor_type	CUSTOMER (QR), STAFF (seated), AI
guest_count	smallint	nullable
guest_phone	varchar(20)	optional, for receipt
started_at	timestamptz	not null
ended_at	timestamptz	
last_activity_at	timestamptz	for idle expiry
close_reason	varchar(40)	PAID, TIMEOUT, MANAGER, WALK_OUT

4.4 Menu

menu_categories

Menu sections.

Column	Type	Constraints / Notes
id	uuid PK	
restaurant_id	uuid FK	
name	varchar(100)	
sort_order	int	
is_active	boolean	

menu_items

Dishes and drinks.

Column	Type	Constraints / Notes
id	uuid PK	
restaurant_id	uuid FK	
category_id	uuid FK	
name	varchar(150)	
description	text	
price	numeric(12,2)	base price
tax_rate	numeric(5,2)	GST percent
image_url	text	
is_veg	boolean	
is_available	boolean	sold-out toggle
prep_time_min	smallint	target preparation time
tags	text[]	spicy, bestseller
sort_order	int	

item_variants

Size or portion options.

Column	Type	Constraints / Notes
id	uuid PK	
item_id	uuid FK	
name	varchar(60)	Half / Full
price	numeric(12,2)	replaces base price

modifier_groups and modifiers

Customisation (extra cheese, spice level).

Column	Type	Constraints / Notes
modifier_groups.id	uuid PK	
modifier_groups.item_id	uuid FK	
modifier_groups.name	varchar(80)	
modifier_groups.min_select, max_select	smallint	selection rules
modifiers.id	uuid PK	
modifiers.group_id	uuid FK	
modifiers.name	varchar(80)	
modifiers.price_delta	numeric(12,2)	added to price

4.5 Orders

orders

One order round within a session.

Column	Type	Constraints / Notes
id	uuid PK	
restaurant_id	uuid FK	
session_id	uuid FK	indexed
table_id	uuid FK	denormalised for speed
order_no	int	per-restaurant daily sequence
status	order_status	default PLACED
subtotal, tax_total, discount_total, total	numeric(12,2)	server calculated
notes	text	
placed_by_type	actor_type	CUSTOMER or STAFF
placed_by_user_id	uuid FK null	when staff placed
placed_at	timestamptz	

order_items

Line items with their own status for kitchen tracking.

Column	Type	Constraints / Notes
id	uuid PK	
order_id	uuid FK	
menu_item_id	uuid FK	
variant_id	uuid FK null	
name_snapshot	varchar(150)	name at order time
unit_price	numeric(12,2)	price at order time
quantity	smallint	at least 1
modifiers_snapshot	jsonb	chosen modifiers and deltas
line_total	numeric(12,2)	
status	item_status	default PENDING
instructions	text	
accepted_at, ready_at, served_at	timestamptz	timing analytics
cancel_reason	text	

4.6 Billing and Payments

bills

Running bill for a session.

Column	Type	Constraints / Notes
id	uuid PK	
session_id	uuid FK unique	one bill per session
subtotal, tax_total, service_charge, discount_total, tip	numeric(12,2)	
grand_total	numeric(12,2)	
amount_paid	numeric(12,2)	default 0
status	bill_status	default OPEN
invoice_no	varchar(30)	unique, sequential for GST
generated_at, paid_at	timestamptz	

payments

Every payment attempt.

Column	Type	Constraints / Notes
id	uuid PK	
bill_id	uuid FK	
method	payment_method	
provider	varchar(30)	razorpay
provider_order_id, provider_payment_id	varchar(80)	unique where not null
amount	numeric(12,2)	
status	payment_status	
idempotency_key	varchar(80)	unique; prevents duplicate charge
received_by_user_id	uuid FK null	for cash
raw_payload	jsonb	webhook body
paid_at	timestamptz	

refunds

Refund records.

Column	Type	Constraints / Notes
id	uuid PK	
payment_id	uuid FK	
amount	numeric(12,2)	
reason	text	
approved_by	uuid FK	manager/owner
status	varchar(20)	
created_at	timestamptz	

4.7 Attention, Cleaning and Logs

table_alerts

Things that need staff attention.

Column	Type	Constraints / Notes
id	uuid PK	
restaurant_id	uuid FK	
table_id	uuid FK	
session_id	uuid FK null	
type	alert_type	
severity	smallint	1 info, 2 warning, 3 critical
message	text	
raised_by	actor_type	
status	alert_status	default OPEN
assigned_to	uuid FK null	waiter
raised_at, acknowledged_at, resolved_at	timestamptz	response-time metrics
escalated	boolean	default false

cleaning_tasks

Cleaning queue.

Column	Type	Constraints / Notes
id	uuid PK	
table_id	uuid FK	
session_id	uuid FK null	
status	task_status	
assigned_to	uuid FK null	cleaner
priority	smallint	raised if table is reserved soon
created_at, started_at, completed_at	timestamptz	
notes	text	

table_status_log

Immutable history of every table state change.

Column	Type	Constraints / Notes
id	bigserial PK	
table_id	uuid FK	indexed
from_status, to_status	table_status	
reason	varchar(80)	
actor_type	actor_type	
actor_id	uuid null	user id if staff
created_at	timestamptz	

audit_logs

Sensitive actions.

Column	Type	Constraints / Notes
id	bigserial PK	
restaurant_id	uuid FK	
actor_id	uuid FK null	
action	varchar(60)	DISCOUNT_APPLIED, REFUND, MENU_PRICE_CHANGED
entity, entity_id	varchar, uuid	
diff	jsonb	before and after
ip, user_agent	text	
created_at	timestamptz	

4.8 Feedback, Waitlist and Notifications (Phase 1.5)

feedback

Guest ratings.

Column	Type	Constraints / Notes
id	uuid PK	
session_id	uuid FK	
rating	smallint	1 to 5
comment	text	
created_at	timestamptz	

reservations and waitlist_entries

Demand management.

Column	Type	Constraints / Notes
reservations.id	uuid PK	
reservations.table_id	uuid FK null	
reservations.name, phone, party_size	mixed	
reservations.reserved_for	timestamptz	
reservations.status	varchar(20)	BOOKED, SEATED, NO_SHOW, CANCELLED
waitlist_entries.id	uuid PK	
waitlist_entries.name, phone, party_size	mixed	
waitlist_entries.joined_at, notified_at, seated_at	timestamptz	
waitlist_entries.status	varchar(20)	WAITING, NOTIFIED, SEATED, LEFT

4.9 Phase 2: AI Camera Tables

cameras

Registered camera sources.

Column	Type	Constraints / Notes
id	uuid PK	
restaurant_id	uuid FK	
floor_id	uuid FK	
name	varchar(80)	
stream_url_encrypted	text	RTSP URL stored encrypted
status	varchar(20)	ONLINE, OFFLINE
last_seen_at	timestamptz	

camera_table_zones

Maps a polygon in the camera image to a table.

Column	Type	Constraints / Notes
id	uuid PK	
camera_id	uuid FK	
table_id	uuid FK	
polygon	jsonb	list of normalised x,y points

vision_events

Detections sent by the AI service.

Column	Type	Constraints / Notes
id	bigserial PK	
camera_id	uuid FK	
table_id	uuid FK null	
event_type	varchar(40)	PERSON_SEATED, TABLE_CLEARED, QUEUE_COUNT, LONG_UNATTENDED
person_count	smallint	
confidence	numeric(4,3)	0 to 1
applied	boolean	true if state engine used it
snapshot_url	text null	optional, privacy controlled
detected_at	timestamptz	

5. Key Indexes and Constraints

Purpose	Definition
Find active session fast	Unique partial index on table_sessions(table_id) where status = 'ACTIVE' (guarantees one active session per table)
Floor map query	Index tables(restaurant_id, floor_id, status)
Kitchen queue	Index order_items(status, accepted_at) where status in PENDING/ACCEPTED/PREPARING
Open alerts	Index table_alerts(restaurant_id, status, raised_at) where status != 'RESOLVED'
Cleaning queue	Index cleaning_tasks(status, created_at) where status in PENDING/IN_PROGRESS
No duplicate payment	Unique on payments(idempotency_key) and payments(provider_payment_id)

Purpose	Definition
Invoice sequence	Unique bills(restaurant_id, invoice_no)
QR lookup	Unique tables(qr_token)
Reports	Index orders(restaurant_id, placed_at), payments(paid_at)

6. Data Retention and Privacy

- Guest phone numbers are optional and stored only for receipts, waitlist notifications and feedback.
- Financial records are retained as required by Indian tax rules; personal data can be anonymised on request.
- Phase 2 stores counts and states, not faces. Snapshots are off by default and auto-deleted after a short retention window.
- Daily encrypted backups with point-in-time recovery; backups tested regularly.